INCH-POUND

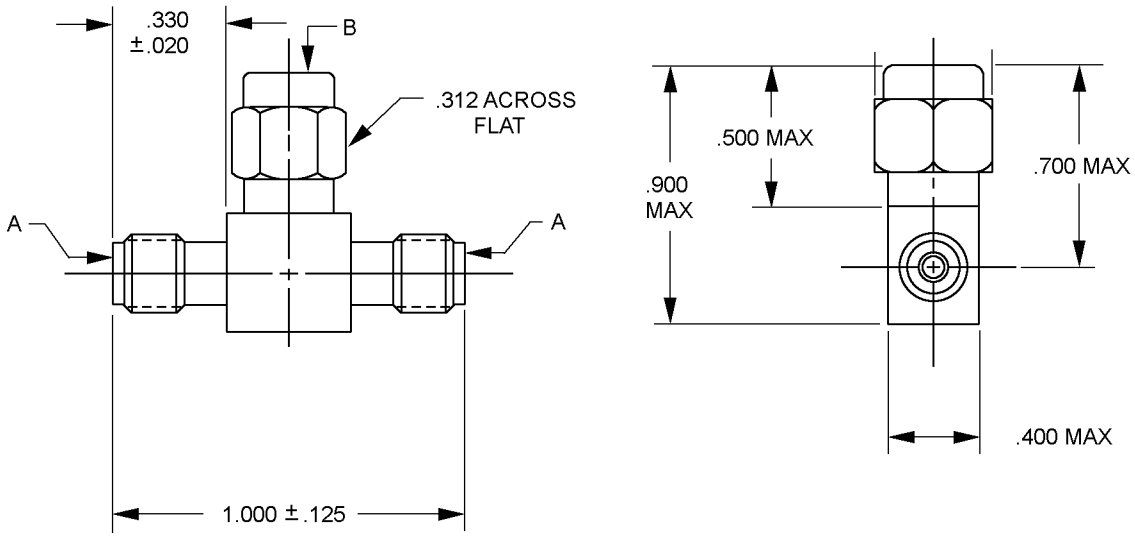
MIL-PRF-55339/30C
w/AMENDMENT 3
28 September 2023
SUPERSEDING
MIL-PRF-55339/30C
w/AMENDMENT 2
6 March 2020

PERFORMANCE SPECIFICATION SHEET

ADAPTER, CONNECTOR, COAXIAL, RADIO FREQUENCY, "T", (WITHIN SERIES SMA PLUG TO TWO SERIES SMA JACKS), CLASS 2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-55339.



M55339/30-30101 without safety wire holes
M55339/30-40101

M55339/30-30001 with safety wire holes
M55339/30-40001

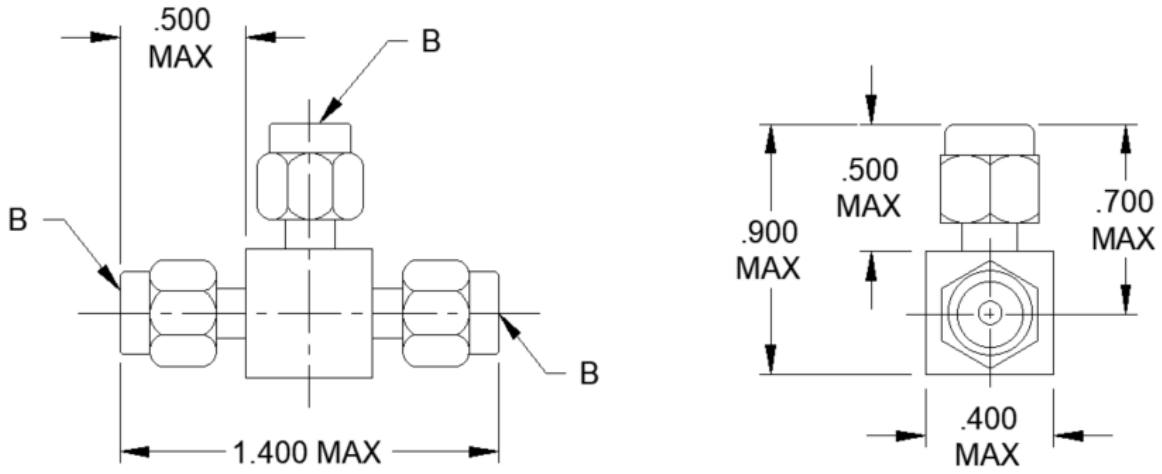
Reference	Series	Contact
A	SMA	Socket
B	SMA	Pin

FIGURE 1. General configuration jack-plug-jack.

AMSC N/A
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

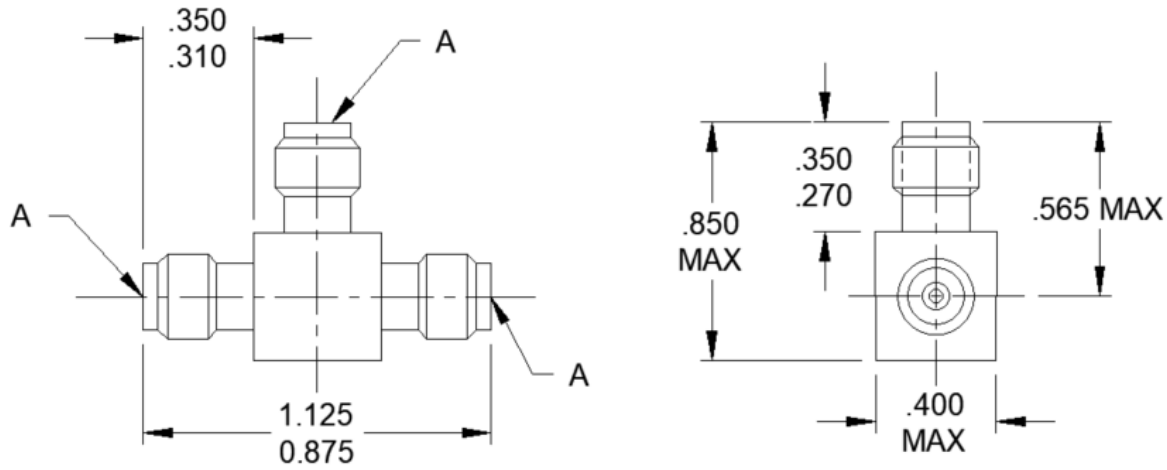
FSC 5935

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Check the source to verify that this is the current version before use.



M55339/30-30102
M55339/30-40102 Without safety wire holes

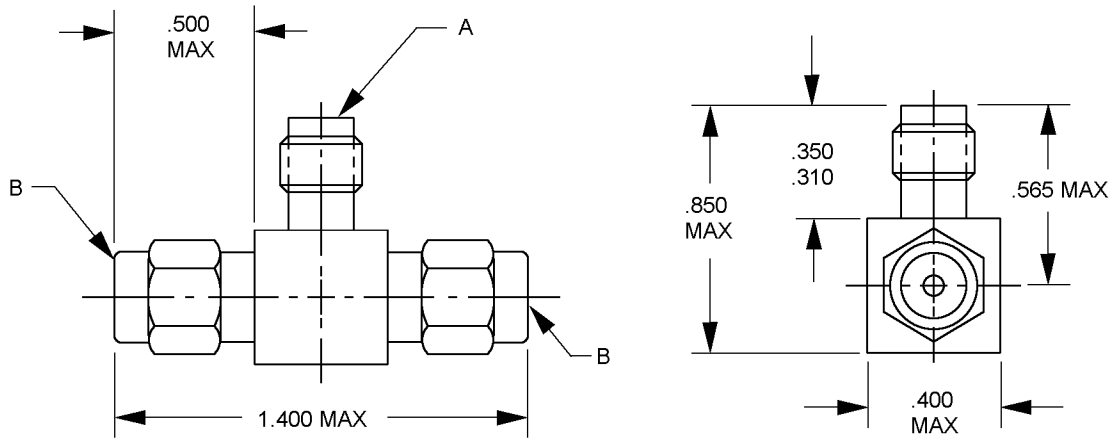
M55339/30-30002
M55339/30-40002 with safety wire holes



M55339/30-30003
M55339/30-40003

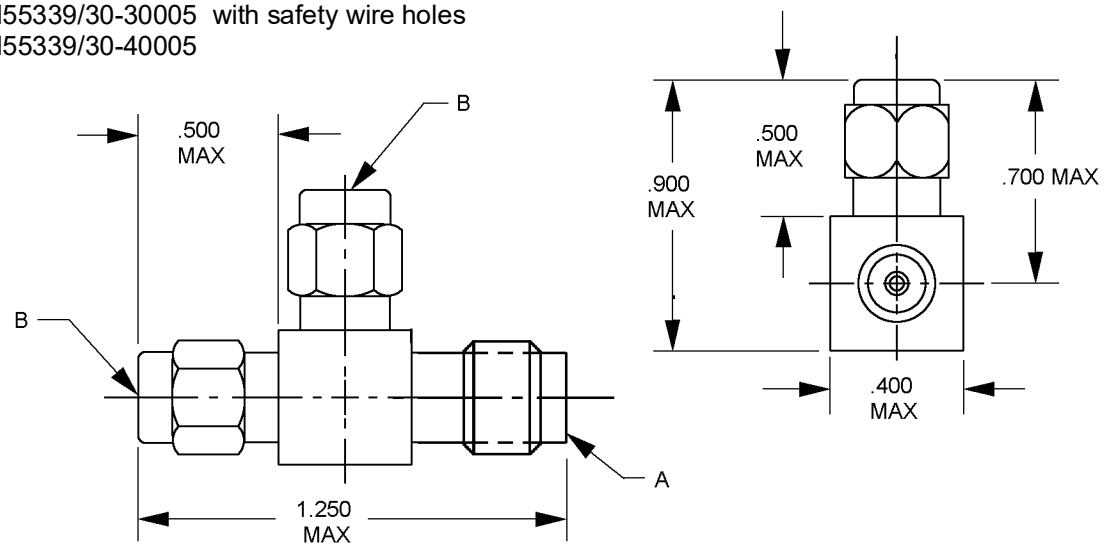
FIGURE 1. General configurations – Continued.

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M55339/30-30105 without safety wire holes
M55339/30-40105

M55339/30-30005 with safety wire holes
M55339/30-40005



M55339/30-30105 without safety wire holes
M55339/30-40105

M55339/30-30005 with safety wire holes
M55339/30-40005

NOTES:
1. All undimensioned pictorial configurations are for reference purposes only.
2. Interfaces shall be in accordance with MIL-STD-348.
3. Three holes .016 (0.41 mm) minimum diameter, equally spaced, are required for safety wiring after mating. Location on coupling nut optional.

FIGURE 1. General configurations – Continued.

ENGINEERING DATA:

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Impedance: 50 ohms, nominal.

Working voltage: Sea level: 335 V rms.
70,000 feet: 85 V rms.

Frequency range: 0 to 12.4 GHz.

Temperature range: -65°C to +165°C.

REQUIREMENTS:

Dimensions and configuration: See figure 1 and section 310 of MIL-STD-348.

Center contact retention:

Axial force: 6 pounds, minimum.
Torque: 4 inch-ounces, minimum.

Inspection conditions. For each test of threaded coupling connectors where the test is performed on mated pairs, the pairs shall be torqued to 7 to 10 inch-pounds.

Force to engage and disengage:

Longitudinal force: Not applicable.
Torque: 2 inch-pounds, maximum.

Coupling proof torque: 15 pounds, minimum.

Mating characteristics:

Center contact (socket):

Oversize test pin diameter: .0375 + .0001 inch, minimum.
Pin finish: 16 microinches.
Insertion depth: .030/.045 inch, minimum.
Number of insertions: Three.

Maximum test pin (insertion force test):
Steel test pin diameter: .0370 + .0001 inch, minimum.
Pin finish: 16 microinches.
Insertion depth: .050/.075 inch, minimum.
Insertion force: 3 pounds, maximum.
Number of insertions: One.

Minimum test pin (withdrawal force):
Steel test pin diameter: .0355 - .0001 inch, minimum.
Pin finish: 16 microinches.
Insertion depth: .050/.075 inch, minimum.
Withdrawal force: 1 ounce, minimum.
Number of withdrawals: One.

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Permeability: Less than 2.0.

Seal: Pressurized: Not applicable.
Weatherproof: Not applicable.

Insulation resistance: MIL-STD-202-302, test condition B. 5,000 megohms, minimum.

VSWR: Not applicable.

RF leakage (total): Not applicable.

RF insertion loss: Not applicable.

Durability: 500 cycles minimum at 12 cycles/minute maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Dielectric withstanding: MIL-STD-202-301.

Test voltage: 1,000 V rms, minimum (sea level).

Contact resistance (milliohms, maximum):

Contact	Initial	After environmental
Center	6.0	8.0
Outer	2.0	Not applicable

Vibration, high frequency: MIL-STD-202-204, test condition D, interruptions – 1 µs, maximum.

Shock (specified pulse): MIL-STD-202-213, test condition I.

Thermal shock: MIL-STD-202-107, test condition C.

Moisture resistance: MIL-STD-202-106. 200 megohms, minimum, within 5 minutes after removal from humidity.

Corona level:

Voltage: 250 V, minimum. Altitude, 70,000 feet, minimum.

RF high potential withstanding voltage:

RF voltage: 670 V rms, minimum.

Frequency: 5 MHz, minimum.

Salt spray (corrosion): MIL-STD-202-101, test condition B.

Coupling mechanism retention force: 60 pounds, minimum.

Group qualification: See table I.

Part or Identifying Numbers (PIN): See figure 1.

Marking: As specified in MIL-PRF-55339.

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NOTE: For logistics purposes, only adapters with safety wire hole will be stocked.

TABLE I. Group qualification.

Group	Submission and qualification of any of the following connectors	Qualifies the following connectors
1	M55339/30-30001	M55339/30-30101 M55339/30-30001 M55339/30-30102 M55339/30-30002 M55339/30-30003 M55339/30-30104 M55339/30-30004 M55339/30-30105 M55339/30-30005
2	M55339/30-40001	M55339/30-40101 M55339/30-40001 M55339-30-40102 M55339/30-40002 M55339/30-40003 M55339/30-40104 M55339/30-40004 M55339/30-40105 M55339/30-40005

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents: In addition to MIL-PRF-55339, this document references the following:

MIL-STD-202-101 MIL-STD-202-213
MIL-STD-202-106 MIL-STD-202-301
MIL-STD-202-107 MIL-STD-202-302
MIL-STD-202-204 MIL-STD-348

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA – CC

Preparing activity:
DLA - CC

(Project 5935-2023-078)

Review activities:
Army – AR, AT
Navy - AS, MC, OS
Air Force – 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of information above using the ASSIST Online database at https://assist.dla.mil.

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